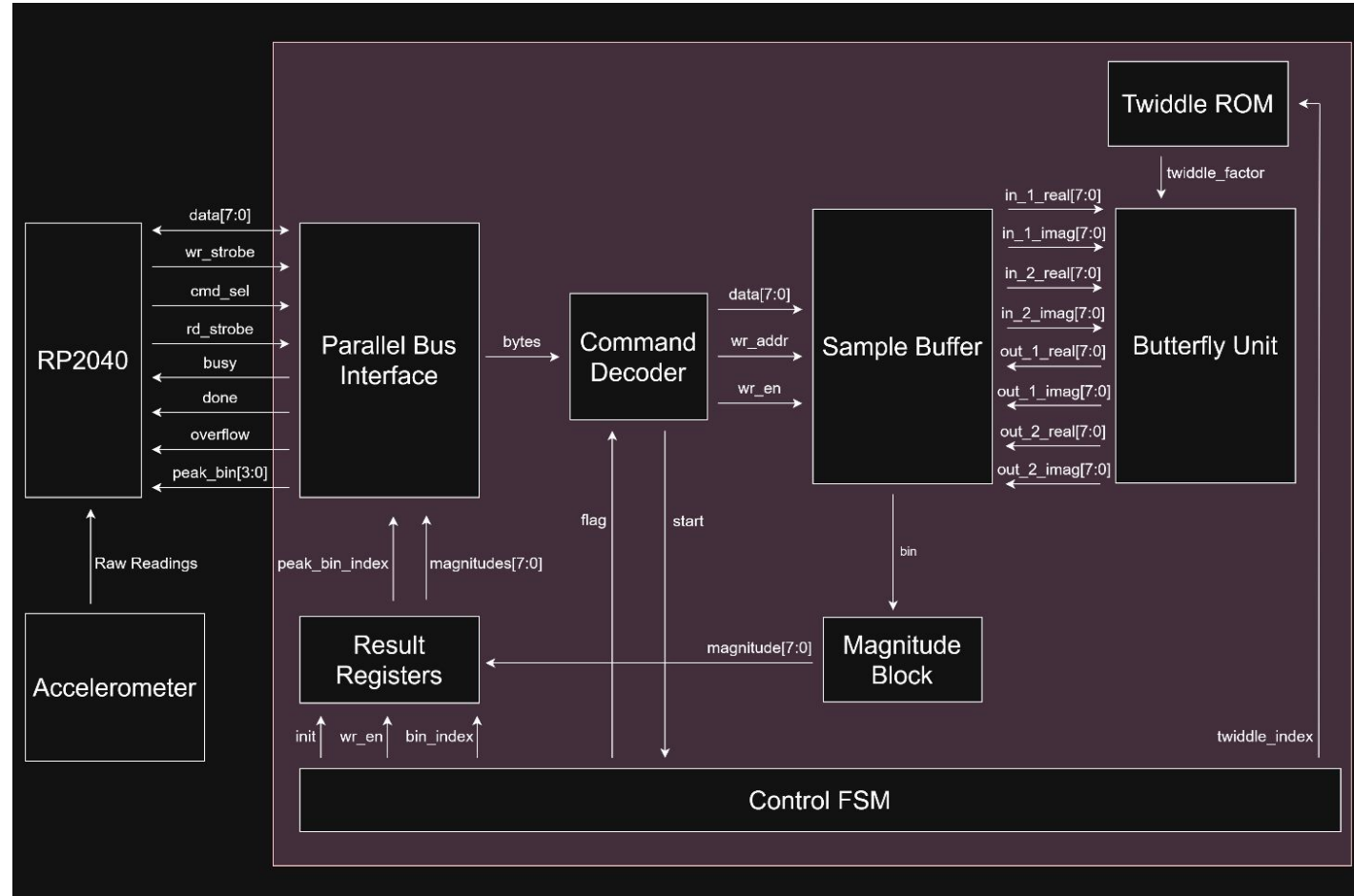


Offloads heavy discrete fourier transform math from main processor, performing math more efficiently than software.

Specifically designed for an **accelerometer**, turning **g-forces** over time to **vibration frequency data**.



### Specifications

- **Clock Freq.:** 20 MHz
- **Latency:**  $\sim 3.55 \mu\text{s}$
- **Clock cycles:**  $\sim 71$
- **FFT size:** 8-point
- **No. of butterfly units:** 1
- **Input data width:** 8 bits
- **Axes Processed:** 3 (x, y, z)

| Pin No. | ui_in     | uo_out      | uio     |
|---------|-----------|-------------|---------|
| 0       | wr_strobe | busy        | data[0] |
| 1       | cmd_sel   | done        | data[1] |
| 2       | rd_strobe | overflow    | data[2] |
| 3       | -         | -           | data[3] |
| 4       | -         | peak_bin[0] | data[4] |
| 5       | -         | peak_bin[1] | data[5] |
| 6       | -         | peak_bin[2] | data[6] |
| 7       | -         | peak_bin[3] | data[7] |

## Task Assignment

- **Eva:**
  - Parallel bus interface
  - Command decoder
  - Result registers
  - Control FSM
- **Fatma:**
  - Magnitude block
  - Sample buffer
  - Butterfly unit
  - Twiddle ROM

| Date         | Task                                                                                        |
|--------------|---------------------------------------------------------------------------------------------|
| Oct 1-4      | Start sample buffer & parallel bus interface.                                               |
| Oct 5-11     | Finish sample buffer & parallel bus.<br>Start twiddle ROM & command decoder.                |
| Oct 12-18    | Finish twiddle ROM & command decoder. Start butterfly unit, result registers & control FSM. |
| Oct 31-Nov 5 | Finish butterfly unit & result registers. Start magnitude block. Continue control FSM.      |
| Nov 6-12     | Finish magnitude & control FSM. Integrate all modules                                       |
| Nov 13-23    | Debug integration, arithmetic & sequencing                                                  |
| Nov 24-27    | Verify output values, commands & debug for edge cases (overflow, negative inputs, etc).     |
| Nov 28-30    | Fix remaining datapath/control issues, write documentation.                                 |
| Dec 1-3      | Review documentation & submit.                                                              |